Domain Mapping Network for Remote Sensing Cross-Domain Few-Shot Classification

Xiaoqiang Lu, *Senior Member, IEEE*, Tengfei Gong, and Xiangtao Zheng, *Senior Member, IEEE*

***Abstract*— It is a challenging task to recognize novel categories with only a few labeled remote-sensing images. Currently, meta-learning solves the problem by learning prior knowledge from another dataset where the classes are disjoint. However, the existing methods assume the training dataset comes from the same domain as the test dataset. For remote-sensing images, test datasets may come from different domains. It is impossible to collect a training dataset for each domain. Meta-learning and transfer learning are widely used to tackle the few-shot classification and the cross-domain classification, respectively. However, it is difficult to recognize novel categories from various domains with only a few images. In this article, a domain mapping network (DMN) is proposed to cope with the few-shot classification under domain shift. DMN trains an efficient few-shot classification model on the source domain and then adapts the model to the target domain. Specifically, dual autoencoders are exploited to fit the source and target domain distribution. First, DMN learns an autoencoder on the source domain to fit the source domain distribution. Then, a target autoencoder is initiated from the source domain autoencoder and further updated with a few target images. To ensure the distribution alignment, cycle-consistency losses are proposed to jointly train the source autoencoder and target autoencoder. Extensive experiments are conducted to validate the generalizable and superiority of the proposed method.**

***Index Terms*— Cross-domain classification, few-shot classification, meta-learning, remote sensing scene classification, transfer learning.**

I. Introduction

REMOTE sensing scene classification focuses on recognizing the scene categories of a remote-sensing image, which provides crucial information for many fields [1], such as urban planning [2], geographic image retrieval [3], and disaster monitoring [4], [5].

An important property of remote-sensing images is the rare samples and annotations [6], [7]. It is difficult to collect and label enough remote-sensing images. Therefore, it is urgently demanded to recognize novel concepts with a few labeled images [8]. Currently, meta-learning solves the problem by learning prior knowledge from another dataset, where the classes are disjoint. It generally assumes the training dataset and the test dataset come from the same distribution (referred to as domain) [9].

For remote-sensing images, the test datasets may come from different domains [10], [11], [12]. The remote-sensing images are usually captured with diverse distribution discrepancies including sensor differences [13], geographic variations [14], and atmospheric effects [15]. Collecting a training dataset for each domain is computationally intensive. To deploy a recognition system in real-world applications, it is essential to recognize images from various domains [16], [17].

This article targets recognizing novel scene categories from various domains with a few labeled images. To address the above challenges, currently, two mainstream methods are proposed: few-shot classification based on meta-learning [18] and cross-domain classification based on transfer learning [22]. The properties of representative methods are summarized in Table I. Few-shot classification tries to recognize new concepts with only a few samples [23]. Meta-learning mimics the human abilities that build connections between new concepts and existing knowledge to allow few-shot classification [24]. A meta-learning model is generally pretrained on a large labeled training set to accumulate meta-knowledge [18]. Typically, a training dataset is collected into episodes simulating the few-shot setting. Model is trained to correctly classify unlabeled images with the provided few labeled images within episodes [25]. It is often assumed that the large training and test datasets come from the same domain [26]. The performance tends to degrade when generalizing to a different domain [27]. This limitation heavily restricts the application in remote sensing scene classification because remote-sensing images vary greatly in distribution due to imaging region changes, seasonal variations, and sensor diversities [28].

TABLE I
METHOD COMPARISON

Methods	Domain Shift	Novel Categories
Few-shot classification [18]	×	√
Fine-tuning [19]	×	×
Domain adaption [20]	√	×
Domain generalization [21]	√	×
Ours	√	√

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Xiaoqiang Lu and Xiangtao Zheng are with the College of Physics and Information Engineering, Fuzhou University, Fuzhou 350108, China (e-mail: xiangtaoz@gmail.com).

Tengfei Gong is with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China.

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Cross-domain classification aims to adapt a model trained on a labeled training set (source domain) to an unlabeled test dataset (target domain). Transfer learning reduces the domain shift to boost the performance on the cross-domain dataset [29]. The typical methods include fine-tuning [19], domain adaptation [20], and domain generalization [30]. 1) Fine-tuning methods refine a model pretrained on the source domain to the target domain [19]. The limitations are the over-fitting tendency on an extremely small dataset [18], [31] and the inefficiency in tackling severe domain shift [32]. 2) Domain adaptation methods reduce domain shift by mapping data points from two domains into a common feature space [20]. 3) Domain generalization methods generalize to new target domains without accessing the target dataset during training [21]. It learns domain universal features from multiple source domains but fails to exploit the specific target domain knowledge [27]. Commonly, these transfer learning methods are limited to recognizing the categories same as the source domain and inefficient in recognizing novel categories [33].

This article tackles a cross-domain few-shot classification task: recognizing novel categories from various domains with a few images. To this end, a domain mapping network (DMN) is proposed. DMN maps the target domain samples to the source domain distribution, and then the few-shot classification model learned on the source domain can work on the target domain images. Due to the limited number of target samples, directly exploiting existing domain adaptation methods may achieve unsatisfactory performance. To this end, we employ an encoder and a decoder for each domain and exploit cycle-consistency losses to help the distribution alignment. To achieve the few-shot learning, a metric-based few-shot classification model graph neural networks (GNNs) [34] is used. GNN learns how to capture the interaction between the unlabeled and labeled images to propagate the label information to the target images.

Specifically, DMN exploits a dual-branch autoencoder and a GNN-based metric function. First, a source autoencoder is employed to fit the source domain distribution and a metric function is learned on the source domain for few-shot classification. Second, a target autoencoder is initiated from the pretrained source autoencoder and then fits the target domain distribution. To ensure adaptation with just a few target images, cycle-consistency losses are proposed to ensure the distribution alignment between the source and target domain. Finally, both the source autoencoder and target autoencoder are jointly trained to implement adaptation effectively. After the adaptation, all target images are fed into the target encoder to extract features. The unlabeled target images are predicted by comparing them with the labeled target images in the few-shot task. The proposed DMN can quickly generalize to different domains, requiring neither multiple source domains [33] nor numerous target images [10], [27].

In summary, the main contributions of this article are listed below.

1) The remote sensing cross-domain few-shot classification is highlighted: recognizing novel categories with only a few labeled images and generalizing well to various domains.
2) A DMN is proposed to cope with the few-shot classification under domain shift, which learns an efficient few-shot classification model on the source domain and then adapts the model to the target domain by dual autoencoders.
3) Cycle-consistency losses are exploited to ensure the alignment between the source and target domains, which performs a quick adaptation without requiring numerous target images.

The rest of this article is organized as follows. Section II reviews the related work. Section III discusses the proposed method in detail. Section IV provides the experiments. Section V concludes this article.

II. Related Work

This section reviews the related work including few-shot classification, domain adaptation, domain generalization, and cross-domain few-shot classification.

A. Few-Shot Classification

The deep learning technique has achieved significant performance in many fields [35], [36], but it generally requires numerous training data. Few-shot classification studies how to quickly adapt the learned model to a novel task with a few examples [23], [37]. Meta-learning has been proposed to solve the challenging few-shot classification settings, which gain experiences by learning from multiple other tasks [24]. The representative methods can be grouped into three subcategories: learn-to-measure, learn-to-adjust, and learn-to-remember.

1) The learn-to-measure method (known as the metric-based method) contains a feature encoder and a metric function [25], [38], which predicts the category by measuring the distances between the unlabeled images and labeled images. The representative methods are MatchingNet [38], ProtoNet [25], RelationNet [23], and GNN [34], [39]. Matching networks (MatchingNets) exploit trainable k-nearest neighbors with cosine similarity for classification and introduce attention and memory mechanisms for rapid learning [38]. Prototypical networks (ProtoNets) apply Euclidean distance and build a prototype representation for each class. Query images are classified by finding the nearest class prototype [25]. Relation networks (RelationNets) implement classification by computing the relation scores between query images and support images, where the relation scores are produced by convolutional neural networks (CNNs) modules [23]. GNNs exploit a graph convolution network as the metric function for the purpose of generic message passing [34]. 2) The learn-to-adjust method learns a meta-learner and then adjusts the meta-learner into a task-specific learner that adapts to the specific few-shot task by a small number of examples. For example, model-agnostic meta-learning (MAML) learns an optimal initial state, which can quickly adapt basic learners to new tasks [18]. 3) The learn-to-remember method stores the previous experience to make accurate predictions after several samples [40], [41]. A representative method is memory-augmented neural networks (MANNs), which process a sequence of data and

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introduce a memory access module focusing on the memory content [41].

B. Domain Adaptation and Domain Generalization

Domain adaptation reduces distribution discrepancy between source and target domains so that the model learned on the source domain can be used on the target domain [16], [22], [29]. The commonly used method maps both source and target domains into a domain-invariant space. For example, Tzeng et al. [20] learned features that are indistinguishable from a domain discriminator. Recently, Lee et al. [42] proposed to disentangle the content and style. The target domain image can be mapped to the distribution of the source domain by transferring the source style into the target content. However, domain adaptation methods generally require a large number of unlabeled target data available during training. It will easily stick in overfitting [31] for few-shot classification, due to the small number of target samples.

Domain generalization aims at generalizing to new unseen domains without accessing the target domain during training [43]. It is achieved by learning a universal model from multiple source domains. A widely used method is splitting the multiple source domains into pseudo-source domains and pseudo-target domains, and then training the model to generalize well on the pseudo-target domain [21]. Recently, Dubey et al. [30] proposed a domain-adaptive method for domain generalization, which uses a few test examples to construct the domain embedding and then learns an adaptive classifier specialized for the current domain.

Both domain adaptation methods and domain generalization methods focus on transferring knowledge among the same categories. However, these methods do not hold for few-shot classification which targets novel categories [27].

C. Cross-Domain Few-Shot Classification

The conventional few-shot classification methods significantly degrade the classification performance when a domain gap exists in the training dataset and test dataset [26], [44]. This motivates the study of few-shot classification cross-domain generalization (widely called cross-domain few-shot classification) [33], [45]. Current methods can be grouped into two groups: domain adaptive few-shot classification [27], and domain generalized few-shot classification [33]. The first group of methods exploits a large amount of data from the target domain to minimize the feature discrepancy between the source and target domains. For example, Dong and Xing [31] exploited adversarial training and sample selecting strategy to learn features unidentifiable for a domain discriminator. Phoo and Hariharan [27] jointly optimized the few-shot classification loss on the source domain and self-supervised loss on the target domain. The second group of methods simulates various feature distributions from multiple source domains and encourages learning representation generalized to different domains [33], [46]. Tseng et al. [33] proposed a feature-wise transformation layer to augment the image features. Zhang et al. [46] proposed to separate illumination features from images and augment samples with these separated illumination features to simulate various illumination conditions (considered as various domains).

III. Method

A. Preliminary on Few-Shot Classification

Few-shot classification task tries to recognize novel categories with only a few labeled images. Formally, for a task with K categories to classify, N labeled images per category (**support set**) provided, few-shot classification classifies the remaining unlabeled images (**query set**) belonging to the K categories, which is termed as a K-way N-shot task. To perform few-shot sets, an existing labeled dataset is used as the training dataset to learn the general information (meta-knowledge). The training dataset $\mathcal{T}^{\text{source}}$ is sampled as mini-batches (episodes) to simulate the test few-shot sets. Specifically, each episode contains a support set and a query set: (i) the support set $\mathcal{S} = \{(\mathbf{s}_i, y_i)\}_{i=1}^{K\times N}$, and (ii) the query set $\mathcal{Q} = \{(\mathbf{q}_j, y_j)\}_{j=1}^{L}$ formed by selecting L different examples from the same K categories.

First, the **meta-train phase** tries to learn the meta-knowledge over multiple episodes. The meta-knowledge is assessed by the performance of a set of tasks (the performance of correctly classifying query images). Specifically, given a training dataset, the meta-knowledge is learned by minimizing the negative log-likelihood loss of the query set given the support set over multiple episodes

$$\mathcal{L}_{\text{FSL}} = \mathop{\mathbb{E}}_{(\mathcal{S},\mathcal{Q})\in\mathcal{T}^{\text{source}}} -\frac{1}{L}\sum_{j=1}^{L}\log p_\theta(y_j|\mathbf{q}_j,\mathcal{S}) \tag{1}$$

where $\mathcal{L}_{\text{FSL}}$ is the few-shot learning loss. $\mathbb{E}$ is the expectation with respect to the training dataset $\mathcal{T}^{\text{source}}$. $p_\theta(y_j|\mathbf{q}_j,\mathcal{S})$ is the probability of query image $\mathbf{q}_j$ belonging to the ground-truth category y_j, predicted by the algorithm parameterized with θ.

Then, the **meta-test phase** adapts the learned model θ to the test task by some means, e.g., further gradient-descent updates [18], or parameterizing an embedding function and recognizing using a metric function [38]. The new parametric model is defined by P and makes a prediction $\hat{y}_j$ for the test query image $\mathbf{q}_j$, formulated as $\hat{y}_j = \arg\max_y P(y|\mathbf{q}_j,\mathcal{S},\theta)$, where y is the label for the support set $\mathcal{S}$ in the test task.

B. Meta-Train for Cross-Domain Classification

This article focuses on recognizing novel categories with a few labeled images (target domain) whose distribution is different from that of the existing labeled dataset (source domain). The existing few-shot classification methods restrict the training dataset and the test dataset sampled from the same domain. However, the performance degrades when applied to a few-shot task from a different domain. This article proposes a DMN to address the challenging task, i.e., mapping the images in the target domain into the source domain distribution. Due to the limited samples in the target domain, instead of directly mapping, cycle-consistency losses are added as regularization by exploiting autoencoders. The framework is shown in Fig. 1. First, the meta-train phase learns a few-shot classification

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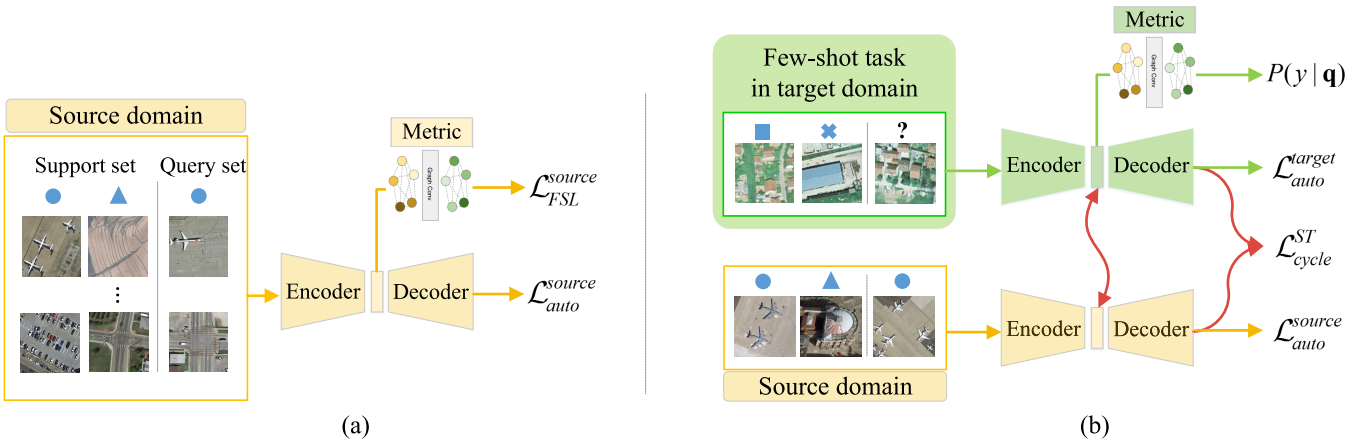


Fig. 1. Illustration of the proposed domain-mapping network. (a) In the meta-train phase, a few-shot classification model is learned on the source domain. Meanwhile, a source autoencoder is exploited to fit the source domain distribution. (b) In the meta-test phase, a target autoencoder is learned to fit the target domain distribution with parameters initiated from the source autoencoder. Cycle-consistency losses are used to retrain the source and target autoencoders jointly. After updating, target images are classified with the target few-shot classification model.

model on the source domain dataset. Specifically, a source autoencoder is employed to fit the source domain distribution with sample reconstruction losses. Then, the meta-test phase fits the target domain distribution by updating a target autoencoder with a few target images, where the target autoencoder is initiated from the source autoencoder. To align the source and target domain, cycle-consistency losses are used to retrain the source and target autoencoders jointly. Finally, unlabeled images are predicted by measuring the distance with the labeled target domain images.

Few-shot classification predicts category by comparing the query image and a few labeled support images with the metric function M. The process in a training episode can be formulated as

$$p_\theta(y_j|\mathbf{q}_j, \mathcal{S}) = M\left(\mathcal{Y}_s^{\text{source}}, E_S\left(\mathcal{Z}_s^{\text{source}}\right), E_S\left(\mathbf{z}_{\mathbf{q}_j}^{\text{source}}\right)\right) \quad (2)$$

where $\mathcal{Y}_s^{\text{source}} = \{y_1, y_2, \ldots, y_{K\times N}\}$ are the labels of the support images. $E_S(\mathcal{Z}_s^{\text{source}})$ denotes source encoder E_S features in support set $\mathcal{S}$. $E_S(\mathbf{z}_{\mathbf{q}_j}^{\text{source}})$ is the source encoder feature in query set $\mathcal{Q}$.

Instead of specifying a fixed similarity, the similarity measure is learned with a parametric model. Specifically, the metric function M exploits a GNN to measure the similarities between query and support images, considering the properties of efficient information prorogation and interaction [34]. Given a task, all images are used to construct a graph. Each node represents a sample. Each edge represents a relationship between two nodes where the similarity is computed using a learnable way. For an L-layer GNN, each GNN layer $\text{Gc}(\cdot)$ receives nodes $\mathbf{x}^{(l)}$ as input and outputs $\mathbf{x}^{(l+1)}$. After a sequence of layers, a fully connected layer following a softmax maps node features to the K-way for category prediction.

The initial node features are the image embedding features concatenated with the one-hot encoding of the label

$$x_i^{(0)} = (\mathbf{f}_i, \mathbf{1}_{y_i}) \quad (3)$$

where $\mathbf{f}_i \in \{E_S(\mathcal{Z}_s^{\text{source}}), E_S(\mathbf{z}_{\mathbf{q}_j}^{\text{source}})\}$ is image feature, and $\mathbf{1}_{y_i}$ is a one-hot encoding of the label $y_i \in \mathcal{Y}_s^{\text{source}}$. For images without labels, $\mathbf{1}_{y_i}$ is the uniform distribution over the K-simplex. The $\text{Gc}(\cdot)$ layer is represented as

$$\mathbf{x}^{(l+1)} = \text{Gc}(\mathbf{x}^{(l)}) = \rho(A\mathbf{x}^{(l)}W^{(l)}) \quad (4)$$

where $W^{(l)}$ are trainable parameters and $\rho(\cdot)$ is a nonlinear activation. Adjacency matrix $A = \{\tilde{A}^{(l)}, 1\}$, where $\tilde{A}^{(l)}$ is learned from the current node representation

$$\tilde{A}_{i,j}^{(l)} = \varphi_{\tilde{\theta}}\left(x_i^{(l)}, x_j^{(l)}\right) = \text{MLP}_{\tilde{\theta}}\left(\text{abs}\left(x_i^{(l)} - x_j^{(l)}\right)\right) \quad (5)$$

where $\varphi_{\tilde{\theta}}$ is a metric parameterized with a multilayer perception $\text{MLP}(\cdot)$. The model is trained to predict labels for the nodes representing query images. Therefore, the final layer is a softmax function that maps the node features to their respective categories $p_\theta(y_j|\mathbf{q}_j, \mathcal{S})$. The model is optimized using the cross-entropy loss defined in (1).

Regarding the time complexity of the GNN propagation in (4), the computation of $\hat{\mathbf{x}}^{(l)} = A\mathbf{x}^{(l)}$ costs $O(\|A\|_0 D)$ in time and $\hat{\mathbf{x}}^{(l)}W^{(l)}$ costs $O(ND^2)$ in time, where D is the feature dimension and N is the number of nodes. Therefore, the total time complexity for the entire network with L layers is $O(L\|A\|_0 D + LND^2)$. In terms of memory complexity, storing the embeddings $\mathbf{x}^{(l)}$ for each layer $l = 1, \ldots, L$ requires $O(LND)$ memory. Both the time and memory complexities are proportional to N, which are suitable for the few-shot classification task with a limited number of images.

The DMN induces a latent feature embedding space to perform domain alignment, which is different from the existing metric-based method. Two autoencoders are leveraged to map source domain images into target domain style and vice versa. Specifically, an encoder denoted as E_S and a decoder denoted as D_S are trained on the source domain for sample reconstruction. The corresponding meta-train loss is

$$\mathcal{L}^{\text{train}} = \mathcal{L}_{\text{FSL}}^{\text{source}} + \mathcal{L}_{\text{auto}}^{\text{source}} \quad (6)$$

where $\mathcal{L}_{\text{FSL}}^{\text{source}}$ denotes the few-shot learning loss $\mathcal{L}_{\text{FSL}}$ trained on the source domain. $\mathcal{L}_{\text{auto}}^{\text{source}}$ represents the autoencoder loss on the source domain.

Given an image I, the feature $\mathbf{z}$ is denoted as $\mathbf{z} = F(I)$, where F denotes the feature extractor. The autoencoder loss

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aims to achieve $\mathbf{z} \approx D_S(E_S(\mathbf{z}))$, which is denoted as

$$\mathcal{L}_{\text{auto}}^{\text{source}} = \mathop{\mathbb{E}}_{(\mathcal{S},\mathcal{Q})\in\mathcal{T}^{\text{source}}} \sum_{\mathbf{z}\in\{\mathcal{Z}_s^{\text{source}},\mathcal{Z}_q^{\text{source}}\}} \times\left\{\lambda_1\|D_S(E_S(\mathbf{z}))-\mathbf{z}\|_1+\lambda_2\,\text{KL}(\{E_S(\mathbf{z})\}||\mathcal{N}(0,I))\right\} \tag{7}$$

where the first term is the reconstruction loss learned over all source domain images $\mathcal{T}^{\text{source}}$, and the second term is the variational loss which promotes the source domain close to the Gaussian distribution with KL divergence. $\mathcal{Z}_s^{\text{source}}$ and $\mathcal{Z}_q^{\text{source}}$ are corresponding image features in the support set $\mathcal{S}$ and query set $\mathcal{Q}$. λ_i are weight parameters.

C. Meta-Test With Few-Shot Adaptation

The meta-test phase adapts the model pretrained on the source domain to the target domain task with the following loss:

$$\mathcal{L}^{\text{test}} = \mathcal{L}_{\text{auto}}^{\text{source}} + \mathcal{L}_{\text{auto}}^{\text{target}} + \mathcal{L}_{\text{cycle}}^{\text{ST}} \tag{8}$$

where $\mathcal{L}_{\text{auto}}^{\text{source}}$ denotes the source autoencoder loss on source domain, $\mathcal{L}_{\text{auto}}^{\text{target}}$ denotes the target autoencoder loss on a few target images, and $\mathcal{L}_{\text{cycle}}^{\text{ST}}$ denotes the cycle-consistency losses between the source domain and target domain.

To fit the target domain distribution with just a few target samples, the target autoencoder is initialized from the source autoencoder, which is composed of the encoder E_T and decoder D_T. The target autoencoder loss is denoted as

$$\mathcal{L}_{\text{auto}}^{\text{target}} = \sum_{\mathbf{z}\in\mathcal{Z}_s^{\text{target}}} \lambda_3\|D_T(E_T(\mathbf{z}))-\mathbf{z}\|_1 + \lambda_4\,\text{KL}(\{E_T(\mathbf{z})\}||\mathcal{N}(0,I)) \tag{9}$$

where the first and second terms are reconstruction loss and the variational loss on the target images. $\mathcal{Z}_s^{\text{target}}$ denotes support image features in the target domain.

To align the distributions between the source and target domain, the following cycle-consistency losses are used:

$$\mathcal{L}_{\text{cycle}}^{\text{ST}} = \lambda_5\mathcal{L}_{\text{STS-cycle}} + \lambda_6\mathcal{L}_{\text{TST-cycle}} + \lambda_7\mathcal{L}_{\text{f-cycle}} \tag{10}$$

$$\mathcal{L}_{\text{STS-cycle}} = \sum_{\mathbf{z}\in\{\mathcal{Z}_s^{\text{source}},\mathcal{Z}_q^{\text{source}}\}} \left\|D_S\left(\overline{E_T}\left(\overline{D_T}(E_S(\mathbf{z}))\right)\right)-\mathbf{z}\right\|_1 \tag{11}$$

$$\mathcal{L}_{\text{TST-cycle}} = \sum_{\mathbf{z}\in\mathcal{Z}_s^{\text{target}}} \left\|D_T\left(\overline{E_S}\left(\overline{D_S}(E_T(\mathbf{z}))\right)\right)-\mathbf{z}\right\|_1 \tag{12}$$

$$\mathcal{L}_{\text{f-cycle}} = \sum_{\mathbf{z}\in\{\mathcal{Z}_s^{\text{source}},\mathcal{Z}_q^{\text{source}}\}} \left\|E_T\left(D_T\left(\overline{E_S}(\mathbf{z})\right)\right)-\overline{E_S}(\mathbf{z})\right\|_1 \tag{13}$$

where $\mathcal{L}_{\text{STS-cycle}}$ denotes the cycle loss from the source domain to the target domain, $\mathcal{L}_{\text{TST-cycle}}$ denotes the cycle loss from the target domain to the source domain, and $\mathcal{L}_{\text{f-cycle}}$ denotes the cycle-encoder loss from source domain to target domain. The line above the autoencoder indicates a parameter frozen during training, which avoids over-fitting to a few target images [47].

IV. Experiments

In this section, we evaluate the performance of the proposed method on the few-shot scene classification tasks. Furthermore, to validate the effectiveness of the domain mapping strategy, a visualization is displayed.

A. Datasets

DMN is trained on a source domain and then assessed on various target domains. To this end, four remote sensing scene classification datasets are used, including NWPU [48], EuroSAT [49], Banja Luka [50], and GID [13]. These datasets are defined as different domains because of collected from different sensors and geographic locations. The largest dataset NWPU is selected as the source domain dataset to ensure adequate prior knowledge learned, and the rest are selected as target domain datasets. To ensure that the categories are novel in the target domains, all overlapping categories are removed. The detailed source/target domain categories are listed in Table II. Some sample images from each domain are displayed in Fig. 2.

1) Source Domain Dataset: NWPU[1] dataset contains 45 scene categories with each category including 700 RGB images of size 256 × 256 pixels. The spatial resolution varies from 0.2 to 30 m per pixel. These images are collected from Google Earth covering more than 100 countries and regions. To construct the training dataset of few-shot classification, the categories disjoint from the target domains are selected. We sample these images into a set of few-shot tasks. The setup of a K-way N-shot task is as follows: randomly sample K categories, then sample N different images from each of the K categories (support set) to input to the model. The model is trained to correctly classify the remaining new images within the K categories (query set).

2) Target Domain Datasets: The model is evaluated on EuroSAT, Banja Luka, and GID datasets, respectively. EuroSAT[2] is a satellite image dataset that consists of 10 categories. There are 2700 RGB images in total, each image is with 64 × 64 pixels. Images are collected from Sentinel-2 satellite spanning 13 spectral bands in the visible, near-infrared, and short-wave infrared parts of the spectrum. Banja Luka[3] dataset is an aerial image dataset containing 606 aerial images divided into six scene categories. Each image is with a size of 128 × 128 pixels in the RGB color space. These images are divided from a larger image (4500 × 6000 pixels), which covers the regions of Banja Luka, Bosnia, and Herzegovina. GID[4] is a satellite image dataset that contains the image-level scene labels and pixel-level land-cover labels. There are 6000 RGB images annotated with scene labels covering 15 categories. Each image has a size of 224 × 224 pixels. The images are obtained from the Gaofen-2 satellite covering a spectral range of blue, green, red, and near-infrared, with a spatial resolution of 4 m multispectral.

[1] http://www.escience.cn/people/JunweiHan/NWPU-RESISC45.html
[2] https://github.com/phelber/eurosat
[3] https://dsp.etfbl.net/aerial/
[4] https://captain-whu.github.io/GID/

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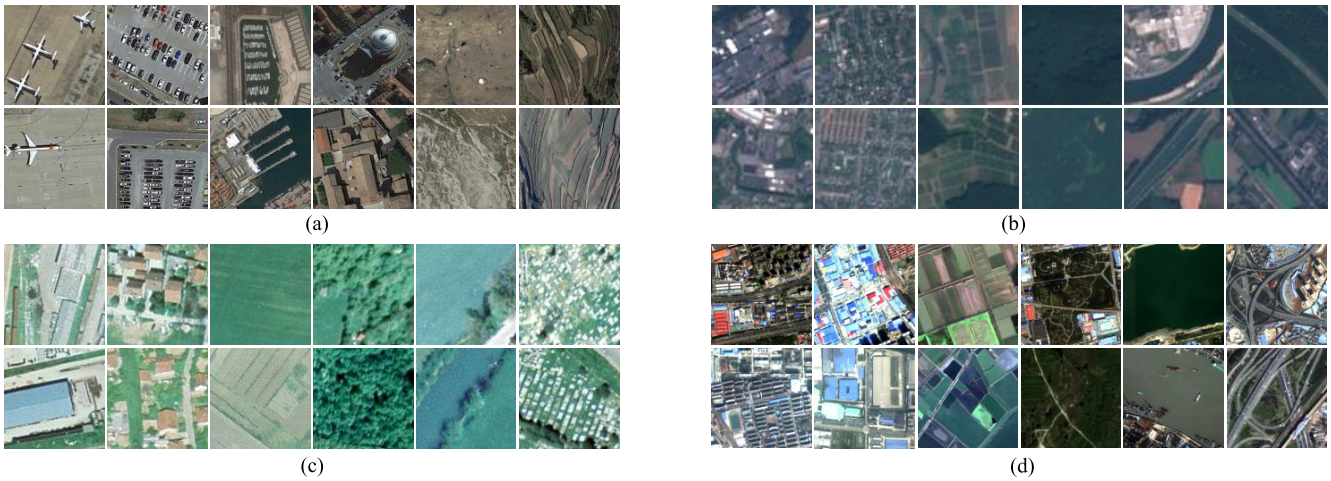


Fig. 2. Some sample images in (a) NWPU, (b) EuroSAT, (c) Banja Luka, and (d) GID datasets. Each dataset displays six categories with two images per category.

TABLE II

DETAILED INFORMATION ABOUT SOURCE DOMAIN (NWPU) CATEGORIES AND TARGET DOMAINS (EUROSAT, BANJA LUKA, AND GID) CATEGORIES

Domain	Dataset	Categories
Source	NWPU (partial)	airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, commercial area, desert, golf course, ground track field, harbor, intersection, island, lake, mobile home park, mountain, palace, parking lot, railway, railway station, roundabout, runway, seaice, ship, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, wetland
Target	EuroSAT (partial)	industry, residential, highway, annual crop, permanent crop, herbaceous vegetation, forest, pasture, river
	Banja Luka	industry, house, field, trees, river, cemetery
	GID (partial)	arbor forest, garden land, pond, shrub land, artificial meadow, industrial land, natural meadow, river, traffic land, dry cropland, irrigated land, paddy field, rural residential, urban residential

B. Implementation Details

This section presents the details of the network architecture, the training and test strategy, as well as the different types of evaluation paradigms.

ResNet-10 is used as the backbone. Specifically, the ResNet-10 [51] is split into two parts which are used as the feature extractor and encoder, respectively. The top part contains a 7×7 convolution layer and two residual blocks. The bottom part contains the remaining two residual blocks. The decoder is constructed to be symmetric to the encoder which also contains two residual blocks except for exploiting transposed convolutions. Features used for classification are the output of the encoder after an average-pooling. A graph convolution module is used as the metric function following the state-of-the-art GNN [34]. $\lambda_1 = \lambda_3 = \lambda_5 = \lambda_6 = 1$, $\lambda_2 = \lambda_4 = \lambda_7 = 0.001$.

In the meta-train phase, the model is trained for 200 epochs with each epoch containing 600 episodes (tasks). The Adam optimizer is used. The learning rate is set to 0.001. In the meta-test phase, for each task in the target domain to be classified, one task from the source domain is sampled. The support set of the target domain task together with the support set and the query set of the source domain task is used to update the source autoencoder and target autoencoder jointly. The model adapts 20 epochs on each test task. The learning rate is set to 0.0002. After the updating, the query set images of the target domain task are classified using the features extracted from the target domain encoder.

Two types of evaluation paradigms are conducted. The first one is constructing the target domain datasets into a set of few-shot tasks, i.e., the widely used K-way N-shot setting [26]. The setting is as follows: select K categories from a target domain dataset, provide the model with N different images for each K category, and evaluate the model's ability to classify new images within the K categories. In the experiments, five-way one-shot and five-way five-shot tasks are considered. The query set is set to contain 15 images per category. Therefore, there are $5 \times 1 + 5 \times 15 = 80$ images used to construct a five-way one-shot task and $5 \times 5 + 5 \times 15 = 100$ images for a five-way five-shot task. Evaluation is conducted over 600 randomly sampled few-shot classification tasks. Average accuracy and 95% confidence intervals are reported.

The second evaluation paradigm is classifying all categories in one dataset, where each category is provided with a few support samples [10]. Specifically, in the K-way N-shot task, the K is set to equal the number of the classes of the target domain. This paradigm can classify all categories at one time. However, the model must be meta-trained from scratch every

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TABLE III

ABLATION EVALUATION WITH RESPECT TO AVERAGE ACCURACY (%) ON THREE DATASETS. (a) ABLATION RESULTS ON FIVE-WAY FIVE-SHOT. (b) ABLATION RESULTS ON FIVE-WAY ONE-SHOT

(a) Ablation results on 5-way 5-shot.

Method	EuroSAT	Banja Luka	GID	Average
GNN	79.14 ± 0.76	97.34 ± 0.21	71.71 ± 0.88	82.73
GNN+Rec	79.53 ± 0.71	97.87 ± 0.19	72.13 ± 0.76	83.18
GNN+Rec+Map	**80.35 ± 0.66**	**98.13 ± 0.18**	**73.32 ± 0.76**	**83.93**

(b) Ablation results on 5-way 1-shot.

Method	EuroSAT	Banja Luka	GID	Average
GNN	59.59 ± 1.09	80.94 ± 0.77	51.03 ± 1.02	63.85
GNN+Rec	61.12 ± 1.14	84.07 ± 0.79	52.13 ± 0.93	65.77
GNN+Rec+Map	**61.91 ± 1.13**	**87.98 ± 0.66**	**52.38 ± 1.02**	**67.42**

time when comes across a new target dataset. Because different target dataset contains a different number of classes.

C. Ablation Experiments

To validate the effectiveness of the domain mapping, we design some ablation settings. Experiments are tested on three datasets. The five-way one-shot and five-way five-shot scenarios are evaluated. The quantitative results are shown in Table III.

The proposed method exploits a metric function proposed in GNN [34], therefore GNN is used as the baseline. GNN + Rec refers to the model updated with loss $\mathcal{L}^{\text{train}}$, i.e., a reconstruction loss employed to train the source autoencoder. The model specified for the source domain is used for the target domain classification task. This variant helps to explore the impact of the reconstruction loss. GNN + Rec + Map refers to the domain mapping exploited, i.e., the full model of the proposed method.

GNN + Rec outperforms GNN which validates that the classification benefits from the reconstruction. It is further observed that GNN + Rec has vast performance improvements in the one-shot setting. It may be attributed to the effect of reconstruction yielding good feature representation. The contribution is significant when the number of support examples is low, but less significant when the number of support examples is high. GNN + Rec + Map further has a performance improvement compared to GNN + Rec. The results suggest a performance forward of the proposed DMN.

D. Compared Methods

There are a few methods of studying cross-domain few-shot classification. Most methods require a target domain or multiple source domains accessible during training which is inapplicable in our setting. Following the recent studies [26], [44], three types of methods are compared: transfer learning-based methods, few-shot classification methods with adaptation, and a state-of-the-art cross-domain few-shot method.

First, it is reported that transfer learning-based methods achieve comparable performances in the cross-domain few-shot classification setting [44]. We, therefore, consider two implementation methods: *Fixed feature extractor (Fixed)* which uses ResNet-10 pretrained on the source domain dataset (with standard supervised learning) as a fixed feature extractor, and learns a new linear classifier on the support set of target domain task to classify the query set; *Fine-tuning all layers (Ft All)* which pretrains ResNet-10 on the source domain dataset and then fine-tunes the pretrained parameters together with learning a new classifier on the support set of the target domain task. Both the Fixed method and Ft All method are updated for 100 epochs on the support set of the target domain task.

TABLE IV

PERFORMANCE COMPARISON WITH RESPECT TO AVERAGE ACCURACY (%) ON THE EUROSAT DATASET

Method	EuroSAT	
	5-shot	1-shot
Fixed	79.27 ± 0.55	59.94 ± 0.88
Ft All	79.31 ± 0.52	59.12 ± 0.79
MAML* [18]	78.04 ± 0.53	52.87 ± 0.86
ProtoNet* [25]	71.49 ± 0.64	52.88 ± 0.86
RelationNet* [23]	64.45 ± 0.61	52.12 ± 0.84
MatchingNet* [38]	79.12 ± 0.53	56.12 ± 0.82
GNN* [34]	79.36 ± 0.64	60.11 ± 0.85
FWT [33]	72.34 ± 0.47	56.84 ± 0.66
DMN (Ours)	**80.35 ± 0.66**	**61.91 ± 1.13**

Second, the strategy of applying adaptation on conventional few-shot classification methods is compared following recent methods [26], [27]. Five state-of-the-art few-shot classification methods are compared: *MAML* [18], *ProtoNet* [25], *RelationNet* [23], *MatchingNet* [38], and *GNN* [34]. To distinguish from the original methods, a star is used to denote that the adaptation strategy is exploited when showing the experimental results in the tables. For the ProtoNet, MatchingNet, and GNN, the model meta-learned on the source domain is used to extract features for a target domain task. The features are then fixed and used to train a new linear classifier as the

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TABLE V
PERFORMANCE COMPARISON WITH RESPECT TO AVERAGE ACCURACY (%) ON THE BANJA LUKA DATASET

Method	Banja Luka	
	5-shot	1-shot
Fixed	92.84 ± 0.27	80.85 ± 0.64
Ft All	93.48 ± 0.30	73.64 ± 0.75
MAML* [18]	90.53 ± 0.35	74.71 ± 0.76
ProtoNet* [25]	94.13 ± 0.26	77.55 ± 0.68
RelationNet* [23]	87.76 ± 0.33	50.87 ± 0.74
MatchingNet* [38]	91.60 ± 0.28	73.34 ± 0.66
GNN* [34]	98.02 ± 0.12	83.55 ± 0.80
FWT [33]	97.80 ± 0.15	83.18 ± 0.55
DMN (Ours)	**98.13 ± 0.18**	**87.98 ± 0.66**

TABLE VI
PERFORMANCE COMPARISON WITH RESPECT TO AVERAGE ACCURACY (%) ON THE GID DATASET

Method	GID	
	5-shot	1-shot
Fixed	72.64 ± 0.65	**53.01 ± 0.84**
Ft All	71.52 ± 0.66	48.46 ± 0.77
MAML* [18]	65.20 ± 0.75	49.28 ± 0.81
ProtoNet* [25]	69.81 ± 0.67	48.59 ± 0.77
RelationNet* [23]	60.25 ± 0.71	46.16 ± 0.84
MatchingNet* [38]	69.50 ± 0.69	48.32 ± 0.77
GNN* [34]	69.66 ± 0.68	50.98 ± 0.85
FWT [33]	65.77 ± 0.54	48.88 ± 0.66
DMN (Ours)	**73.32 ± 0.76**	52.38 ± 1.02

above transfer strategy. For MAML, since it is an initialization method and impractical to fix the features, we update the model to adapt to the target domain tasks with 100 epochs instead of the original one gradient step. For RelationNet, since the outputs are feature maps instead of feature vectors, it is infeasible to connect a linear classifier. Following [26], on the five-way five-shot setting, the five-shot labeled images in a target domain task are split into three pseudo-support and two pseudo-query data to fine-tune the relation module for 100 epochs. On the five-way one-shot setting, no adaptation is employed because the one-shot sample cannot be further split.

Finally, for the cross-domain few-shot classification method, feature-wise transform (FWT) [33] is compared because it is the only method similar to our experimental setting. FWT modulates features with scaling and bias parameters to mimic various distributions so as to generalize to unseen domains. It includes two ways of determining the parameters: learning from multiple source domains or empirically setting fixed parameters (0.3, 0.5) when one source domain is available. To be consistent with the setting of our benchmark, the fixed parameters are adopted.

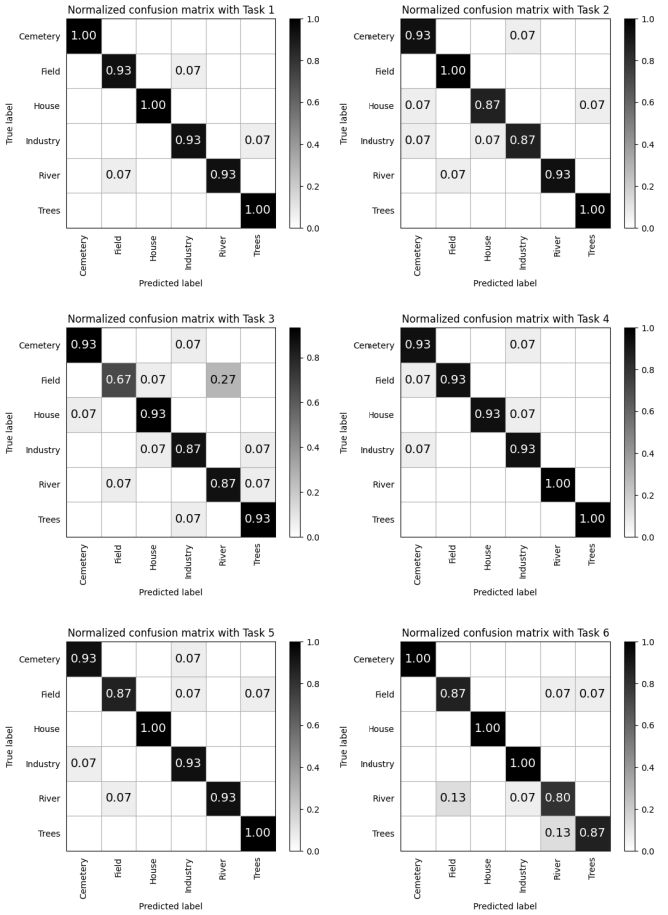


Fig. 3. Confusion matrix of classification results for six different tasks sampled on the Banja Luka dataset.

E. Results Analysis

1) Results on Five-Way Paradigm: The five-way five-shot and five-way one-shot tasks are evaluated. Experiments are conducted on EuroSAT, Banja Luka, and GID. Comparison results with state-of-the-arts are shown in Tables IV–VI, respectively.

It can be seen that the proposed method achieves comparable performance in most scenarios. The transfer learning-based methods perform better than most single-domain few-shot methods, whilst our method further outperforms transfer learning-based methods. FWT exploits fixed hyper-parameters for different domains, which may be unsuitable for remote sensing datasets, thus achieving inferior performances.

In addition, the proposed method has a relatively stable high performance, which further shows its superiority. Taking the GNN* as an example, which achieves the second highest accuracy in the EuroSAT dataset of the five-way five-shot task, it has a significant accuracy drop on the GID dataset of both the five-way five-shot and five-way one-shot tasks. The same is true for the method Fixed and some other methods. However, our method performs well on most datasets.

On the EuroSAT and Banja Luka datasets, our method outperforms other methods on both the five-way five-shot and the five-way one-shot settings. On the GID dataset, our method is inferior to Fixed on the five-way one-shot setting. The reason may be that the GID dataset has many

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TABLE VII
PERFORMANCE COMPARISON OVER ALL CATEGORIES ON THE BANJA LUKA DATASET (%)

Class	Cemetery	Field	House	Industry	River	Trees	OA	Kappa
Fixed	96.93	68.47	99.07	91.40	92.00	96.13	90.67	89.43
Ft All	95.87	84.07	98.00	93.13	95.87	91.67	93.10	92.79
ProtoNet* [25]	95.67	71.20	98.47	90.80	87.47	91.67	89.21	87.24
GNN* [34]	95.00	74.07	97.13	93.13	87.00	95.40	90.29	89.23
FWT [33]	98.93	85.19	96.91	95.13	86.19	97.83	93.36	92.20
DMN (Ours)	95.20	90.87	97.07	96.07	89.20	98.20	**94.43**	**93.33**

TABLE VIII
MCNEMAR'S TESTS ON STATISTICAL SIGNIFICANCE BETWEEN THE DIFFERENT METHODS ON THE BANJA LUKA DATASET

	χ^2	p-value
Fixed vs DMN	40.76	<0.0001
Ft All vs DMN	12.89	0.0003
ProtoNet* vs DMN	46.92	<0.0001
GNN* vs DMN	42.60	<0.0001
FWT vs DMN	4.21	0.0402

images partially overlapped within a category, resulting in an extremely similar support set and query set. A fixed model is good enough for the classification when the supported image is very limited (such as the one-shot). But when the support images increase, the model can have more samples to update to improve performance. Overall, the high performance of DMN on other datasets validates the generalizability of our method. Furthermore, it is observed that the proposed method has large performance gains compared to other few-shot methods, especially in the five-way one-shot setting. It may be attributed to the effect of domain mapping versus fine-tuning. The DMN takes effect in reducing distribution shift with extremely few images (one-shot) while fine-tuning works better when there are a large number of support images. It further demonstrates the effectiveness of the proposed method on the few-shot classification task.

2) Results on All Categories Classification: We classify all categories in the target domain dataset at one time, and report the overall accuracy (OA), kappa coefficient, and classification accuracy of each class, following the setting in the article [10].

Considering the large number of categories on the EuroSAT and GID datasets, and the relatively poor performance of few-shot classification tasks on a large number of categories, the experiments are conducted on the Banja Luka dataset only. The support number N for each task is set to five. Since the choice of different support samples will lead to completely different results, we averaged over 100 tasks. The results are shown in Table VII. In addition, the confusion matrix is displayed in Fig. 3. Since the classification results are different with different support samples, we visualized the results of six different tasks. It is observed that the proposed method outperforms other comparison methods and has a relatively stable performance on different classes.

3) Test for Statistical Significance: To validate the statistical significance, we run McNemar's chi-square test (with Yates' continuity correction) to compare DMN with other competitors. The results are shown in Table VIII. Each row displays the χ^2 and p-value of a test between DMN and one of the other methods. Experiments are conducted on the Banja Luka dataset with the six-way five-shot task. The null hypothesis is that the improvements are due to chance. From Table VIII, we see that the χ^2 values are greater than the critical value (at 95% significance level) of 3.84 and the p-values are less than the significance level (α) of 0.05. Thus, the null hypothesis can be rejected and we can conclude that the improvements obtained with our methods are statistically significant.

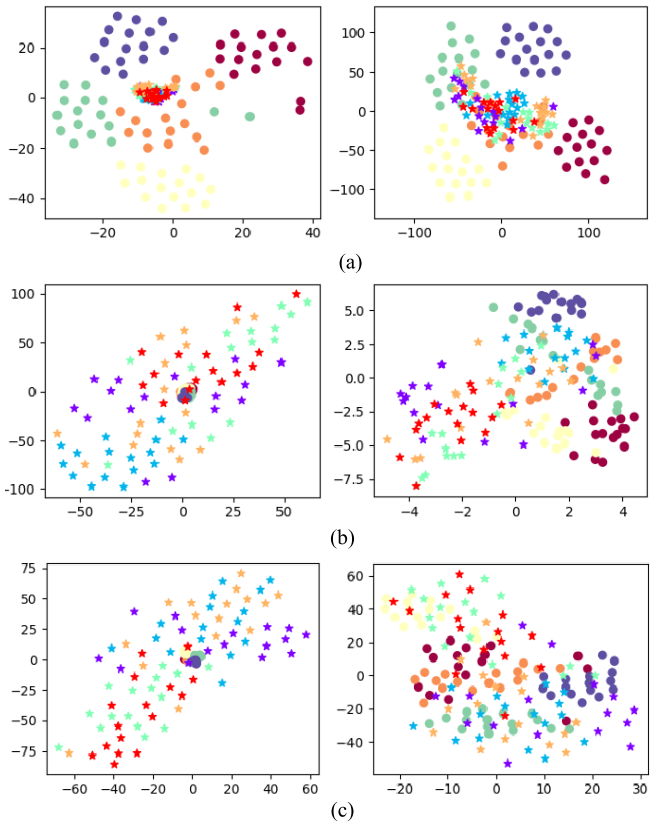


Fig. 4. t-SNE feature visualization of features before (left) and after (right) domain mapping. The circle denotes features from the source domain, and the star denotes features from the target domain. Different colors represent different categories. (a) NWPU→EuroSAT, before and after. (b) NWPU→Banja Luka, before and after. (c) NWPU→GID, before and after.

4) Visualization Results: We present visual results to validate the effectiveness of the DMN. The visualization is conducted from the feature level and image level.

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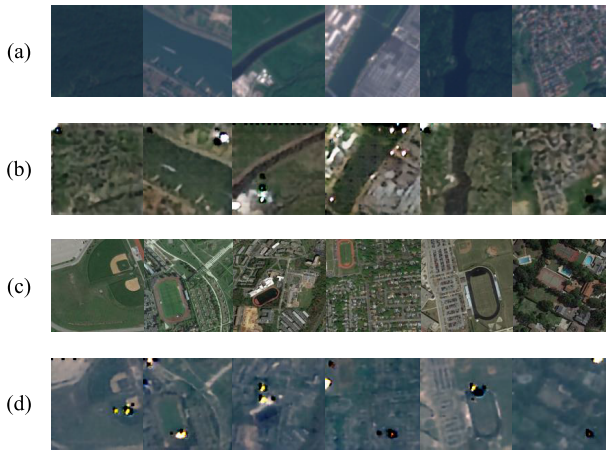


Fig. 5. Domain mapping visualization at the image level. (a) Real images of the target domain (EuroSAT). (b) Generated images with the source domain style. (c) Real images of the source domain (NWPU). (d) Generated images with the target domain style. The top and second rows denote mapping target images to the source domain style. The third and bottom rows denote mapping source images to the target domain style.

First, Fig. 4 displays the features before and after domain mapping. The experiments are conducted on three pairs of cross-domain datasets, which are denoted as NWPU → EuroSAT, NWPU → Banja Luka, and NWPU → GID, respectively. For each domain, one few-shot task is sampled. Each task is composed of one support image and 15 query images selected from five categories. The features are the output of the corresponding domain encoder. t-SNE is exploited to map the features into 2-D space [52]. It is observed that the features of the two domains are projected to a common feature space after mapping. This indicates that the DMN helps the alignment between the source and target domains.

Second, Fig. 5 displays learned images after domain mapping. Experiments are conducted on remote-sensing images of the source domain dataset NWPU and the target domain dataset EuroSAT. The mapping is conducted over the five-way one-shot setting. It can be seen that the DMN maps images from one domain to another style successfully, i.e., accomplishes domain transformation to an opposite domain while ensuring the feature conservation of its own domain, which validates the effectiveness of the proposed method.

V. Conclusion

This article aims to learn a generalized few-shot model for remote sensing scene classification: recognizing novel categories from various domains with a few labeled images. A DMN is proposed to address domain shift under the few-shot setting. Specifically, dual autoencoders are exploited to map images in the target domain into the source domain distribution. In the meta-train phase, a source autoencoder is trained to fit the source domain distribution. In the meta-test phase, a target autoencoder is employed and jointly trained with the source autoencoder to map the source domain and target domain into a common feature space. Extensive experiments on different domains are conducted to verify the effectiveness of the proposed method. In the future, we will further explore the effectiveness of the proposed method on the hyperspectral remote sensing data and the performance of few-shot classification on a large number of categories.

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Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His research interests include intelligent optical sensing, pattern recognition, machine learning, and hyperspectral image analysis.



Tengfei Gong received the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2022.

She is currently a Lecturer with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China. Her research interests include pattern recognition, computer vision, and machine learning.



Xiangtao Zheng (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees in signal and information processing from the Chinese Academy of Sciences, Xi'an, China, in 2014 and 2017, respectively.

He is currently a Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His main research interests include computer vision and pattern recognition.

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